(Please write your Enrollment Number) Enrollment No._____

MID-TERMEXAMINATION
(CBCS)
Course Name: B.Tech (IT/CSE-AI), Semester: 5
(Oct, 2022)

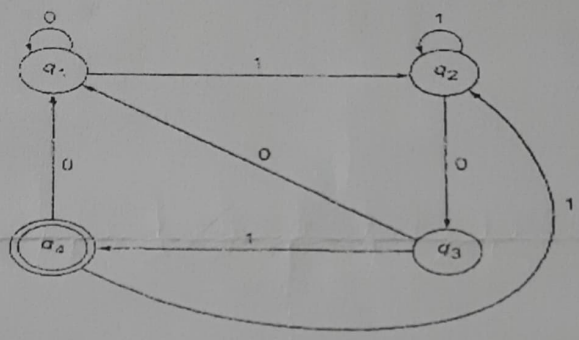
Subject Code: BCS 303 **Subject: Theory of Computation**
Time : 1 ½ Hours **Maximum Marks : 30**
Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) Construct a DFA that accepts the strings over {0, 1} whose decimal representation is divisible by 5.
(b) Write Regular expression forcomplement of L where
$L = \{a^n b^m : n \geq 3, m \leq 4\}$
(c) Construct Context Free Grammar for the language
$L = \{a^n b^m \mid n + m \text{ is even}\}$
(d) Find the reduced grammar equivalent to the grammar S→ aAa, A→ bBB, B→ ab, C→ aB.

Q2 Both Parts are compulsory, Each carry 5 marks each. **(5,5)**

(a) Obtain regular expression for the following finite automata using Arden's theorem. Here q1 is the initial and q4 is final state.



(b) Construct a NDFA equivalent to the Regular Expression **(0 + 1)* (00 + 11) (0 + 1)*** and then convert it into a DFA with reduced states.

Q3 Both Parts are compulsory, Each carry 5 marks each. **(5,5)**

(a) Reduce the following CFG into Chomsky normal form
S→ ASB, A→ aAS | a | ϵ,B→SbS | A | bb
(b) Construct the GNF for the following grammar
S→AB, A→BS|b, B→SA|a

Q4 Both Parts are compulsory, Each carry 5 marks each. **(5,5)**

(a) Design an optimized DFA for the following language
$L = \{w_1 a w_2 : |w_1| \geq 3, |w_2| \leq 4\}$ and $\sum=\{a,b\}$.

(b) Explain the leftmost and rightmost derivation with a suitable example. For the string 'baababaa' check whether the following grammar is ambiguous or not.

S→ a | Sa | bSS | SSb | SbS